

Can Signal Ensembles Beat the S&P 500?

An Empirical Study of Cross-Sectional Predictability, Transaction Costs, and
Machine-Learning Meta-Models

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Abstract

We investigate whether machine-learning models applied to cross-sectional stock characteristics can construct a U.S. equity strategy with a higher Sharpe ratio than the S&P 500. We study two approaches. First, using CRSP daily data (2000–2024), we build a technical-signal ensemble (reversal, momentum, volatility) combined by an XGBoost meta-model and find that the gross profitability is a micro-cap bid–ask–bounce artifact that collapses on a liquid, tradable universe. Second, using monthly CRSP returns merged with Compustat fundamental characteristics (profitability, growth, R&D intensity), we train Lasso, gradient-boosted trees, and a feed-forward neural network to predict the cross-sectional rank of next-month returns. On the full universe, the monthly models extract predictive signal ($IC \approx 0.09$) but fall short of the S&P 500 Sharpe. When restricted to the smallest firms by revenue, XGBoost delivers a net Sharpe of 1.12 (vs. S&P 0.77), remaining above the benchmark for trading costs up to ≈ 57 bps, which is within the range of institutional small-cap estimates. However, this strategy does experience high volatility and large maximum drawdowns. At the daily horizon, all model classes yield indistinguishable predictive content, while at the monthly horizon non-linear models outperform linear ones. Performance is regime-dependent, with the S&P 500 Sharpe rising from 0.19 (2000–2018) to 0.76 (2019–2024).

1 Introduction

The cross-section of stock returns exhibits well-documented empirical regularities. Short-term reversal, momentum, and the low-volatility anomaly among them have motivated decades of factor-based investing. A natural modern question is whether combining several such signals through a machine-learning model can produce a strategy whose risk-adjusted performance exceeds that of a passive benchmark. Gu et al. (2020) show that machine-learning methods meaningfully improve return prediction relative to linear models; yet a persistent gap separates statistical predictability from a *tradable* strategy, because transaction costs and the limited liquidity of anomaly stocks erode paper returns.

In the first stage we use CRSP daily data to engineer and validate technical signals (reversal, momentum, volatility). This reveals a fundamental identification problem: the strongest signals concentrate in illiquid micro-cap stocks where the gross return is dominated by the bid–ask bounce rather than alpha. In the second stage we pivot to a monthly frequency and merge CRSP returns with Compustat firm fundamentals to build a wider predictor set that includes profitability margins, earnings quality, and investment characteristics.

Throughout, we report results net of realistic transaction and borrowing costs against an explicitly defined trading universe¹. Without this discipline, apparent profitability collapses into untradable micro-cap effects. We also treat model choice as an experimental variable, comparing

¹Note that bid–ask spread or price impact are not included, but can be easily added to the cost parameter in the code

an equal-weight blend, gradient-boosted trees (XGBoost), a feed-forward network (MLP), and a recurrent network (LSTM) on the same prediction task at both frequencies.

The findings of this report are:

- At the daily horizon, no strategy beats the S&P 500 net of costs on a liquid universe. All four model classes yield essentially identical predictive content ($IC \approx 0.021$), indicating that the signal, not the model, is the binding constraint.
- Short-term reversal, the strongest daily signal ($IC = -0.054$), owes almost all of its gross profitability to micro-cap bid–ask bounces. On a tradable universe, its edge collapses, which is consistent with reversal as compensation for liquidity provision (Nagel, 2012).
- Combining Compustat fundamentals with technical characteristics yields cross-sectional predictability ($IC \approx 0.09$) at the monthly horizon. XGBoost on the smallest-revenue firms beats the S&P 500 Sharpe for trading costs up to ≈ 57 bps per round trip.
- Performance is regime-dependent: the S&P 500 Sharpe rose from 0.19 over 2000–2018 to 0.76 over 2019–2024, making any systematic strategy that diversifies away from mega-cap names face a high benchmark in the recent period.
- Model sophistication only matters at the monthly horizon. Non-linear models (XGBoost, MLP) outperform linear Lasso by 0.16–0.18 in net Sharpe, whereas at the daily horizon the four model classes share a near-identical $IC (\approx 0.021)$ and none is profitable net of costs.

2 Data and Preprocessing

Data provenance and compliance. All primary data come from the course-provided datasets: the Daily and Monthly CRSP return files, the quarterly Compustat firm-characteristics file, and the Chen–Zimmermann (CZ) anomaly portfolios (Chen and Zimmermann, 2022). Compustat is merged with CRSP through the CRSP/Compustat Merged (CCM) linking table. Two supplementary inputs, the CBOE VIX index and CRSP market capitalisation (price $\times$ shares outstanding), are *not* part of the provided files; they are used *only* for the robustness analyses explicitly flagged in the text (the VIX regime check in Section 3.4 and the market-cap-defined “tradable” universes in Sections 4.3, 4.4 and 6). The headline monthly result (Section 5) uses provided data exclusively, proxying firm size by Compustat trailing-twelve-month revenue. All code is Python, using pandas, NumPy, polars, scikit-learn, XGBoost, PyTorch, statsmodels and Matplotlib.

CRSP returns. We use the Daily CRSP file (2000–2024) and the Monthly CRSP file (sampled from 1980 onward, where Compustat coverage is dense), restricted to ordinary common shares (share codes 10 and 11) on NYSE, AMEX, and NASDAQ. After removing missing returns, clipping extreme returns, and requiring a minimum history per security (based on the features rolling windows), the daily universe peaks near 7 400 names in the early 2000s and declines to $\approx 4 000$ by 2024. The S&P 500 total-return index (`sprtrn`) is the benchmark throughout.

Compustat fundamentals. The provided quarterly Compustat file contains income-statement and cash-flow items in fiscal-year-to-date (YTD) form. We de-cumulate YTD flows to single-quarter figures and aggregate to trailing twelve-month (TTM) totals, matched to CRSP via CCM (`gvkey`→`PERMNO`, 8-digit CUSIP) and lagged four months relative to the fiscal quarter end for point-in-time availability, following Fama and French (2015).

Evaluation methodology and metrics. Predictive accuracy is measured by the information coefficient (IC): the mean per-period cross-sectional Spearman rank correlation between the model score and the realised next-period return. Economic value is measured by the annualised Sharpe ratio of a decile long-short portfolio, *net* of a transaction-cost model (round-trip cost in bps $\times$ portfolio turnover) plus a 1% yr^{-1} borrow charge on the short leg. We also report annualised return, volatility and maximum drawdown, benchmarked against the S&P 500. All splits are strictly chronological. The daily study trains on ≤ 2015 , validates on 2016–2018 and tests on 2019–2024. The monthly study trains on ≤ 2010 , validates on 2011–2014 and tests on 2015–2024. The test set is evaluated once.

3 Signal Construction and Validation

3.1 Feature Engineering

Technical characteristics (5 features). From the monthly CRSP panel we construct: short-term reversal (**reversal_1m**), intermediate momentum (**mom_12_1**, cumulative log-return over months $t-12$ to $t-2$) and short momentum (**mom_6_1**, months $t-6$ to $t-2$) following Jegadeesh and Titman (1993), long-term reversal (**ltr_36_13**, months $t-36$ to $t-13$), and trailing 12-month return standard deviation (**vol_12m**). The prediction target is the next-month cross-sectional return.

Fundamental characteristics (11 features). From Compustat TTM aggregates we derive four profitability margins (operating, gross, EBITDA, net); cash-flow accruals (Sloan, 1996); year-on-year sales growth and earnings change; and four intensity ratios (R&D, capex, SG&A, dividends to revenue). All features are rank-normalised to $[-0.5, +0.5]$ cross-sectionally per holding period.

3.2 Information Coefficient Analysis

We first examine the five technical signals at the daily frequency. Table 1 reports the mean daily cross-sectional Spearman IC for the five technical signals over 2000–2024. Short-term reversal is by far the dominant signal (IC = -0.054 , $t = -35$): a stock’s one-day return negatively predicts the next day’s cross-sectional rank, consistent with Jegadeesh (1990) and Lehmann (1990). Volatility signals carry a stable negative IC (≈ -0.024), the low-volatility anomaly (Baker et al., 2011). Cross-sectional momentum is economically negligible at this frequency (IC $\approx +0.005$).

Table 1: Mean daily cross-sectional rank IC (Spearman) for technical signals, 2000–2024 (6 000–6 300 daily cross-sections). All values are statistically significant at the 1% level except **mom_126d**.

Signal	Mean IC	t -stat	Interpretation
log_reversal_1d	-0.054	-35.2	Short-term reversal
vol_21d	-0.024	-11.3	Low-vol anomaly (short-run)
vol_252d	-0.023	-11.1	Low-vol anomaly (long-run)
mom_21d	-0.018	-11.8	1-month reversal effect
mom_126d	$+0.005$	$+3.2$	Weak 6-month momentum

3.3 Signal Correlation and Effective Dimensionality

The cross-sectional correlation matrix across the 14 daily technical features (including vol-weighted momentum variants) shows approximately three effective families: momentum features are highly intercorrelated ($\rho \approx 0.90$ – 0.93); volatility features form a second cluster; and short-term reversal is

the sole orthogonal signal. The redundancy among technical signals motivates adding orthogonal predictors, which is why we bring in Compustat fundamentals in Section 5.

3.4 Regime Layer: CZ Factors

We test the CZ factors as a potential market-timing overlay. As none forecasts the aggregate market, we exclude them from portfolio construction. We also test the predictive power of the VIX indicator on the market returns and market realized volatility. We conclude that the VIX has an incredible predictive power on the volatility and that it should be considered as a signal for market regime, but not for the direction (OOS $R^2 = 0.41$ when regressing on realized market volatility).

4 Daily Strategy Construction

4.1 Pipeline v1: XGBoost Meta-Model on Technical Signals

We construct a daily long-short decile portfolio: XGBoost ranks all stocks using the five technical features; the top decile is held long and the bottom decile short in equal weight. Transaction cost is charged on daily turnover at 10 bps per round trip; a 1% per year borrow cost applies to the short leg. Over the 2019–2024 test window the gross Sharpe is +4.07, but daily rebalancing generates annual costs of $\approx 81\%$ of the gross return, yielding a net Sharpe of -1.23 against S&P 500 +0.76. XGBoost feature importance assigns 74% weight to reversal. The strategy is effectively a daily reversal book.

4.2 Rebalancing Frequency and Turnover

Across rebalancing frequencies, weekly is the turnover sweet spot for the reversal signal (cost falls from $\approx 85\%/yr$ to $\approx 18\%/yr$), yielding a net Sharpe of +1.01 on the full CRSP universe—above the S&P 500. Low-volatility and momentum exhibit negative gross Sharpe at every frequency during 2019–2024, a regime effect addressed in Section 6.

4.3 Liquidity Filter: Reversal as a Micro-Cap Artifact

The +1.01 weekly reversal Sharpe is computed on the full CRSP universe, which is dominated by illiquid micro-cap names. We apply a tradable-universe filter using supplementary CRSP market-capitalisation data, restricting to the top 500 or top 1 000 stocks.² As Figure 1 shows, reversal’s gross Sharpe collapses from +2.16 (full, weekly) to 0.6–0.8 on the liquid universe, and net Sharpe is negative in every tradable subset (best case: -0.12 , top-1 000, weekly). This result is consistent with Nagel (2012): daily reversal compensates for liquidity provision and is not exploitable by a price-taking investor.

4.4 Long-Only Low-Volatility

As an alternative benchmark, we construct a long-only portfolio tilted toward the lowest-volatility decile within the top-1 000 universe, rebalanced monthly at 10 bps. The 2019–2024 net Sharpe is +0.53, below the S&P 500 at +0.76, because mega-cap technology names are precisely the high-beta stocks that a low-volatility strategy underweights in a concentrated bull market. Regime performance is examined in Section 6.

²Market capitalisation (price $\times$ shares outstanding) is a supplementary input pulled separately from CRSP; the provided daily file contains returns only (see the data note in Section 2). It is used only to define the daily tradable universe.

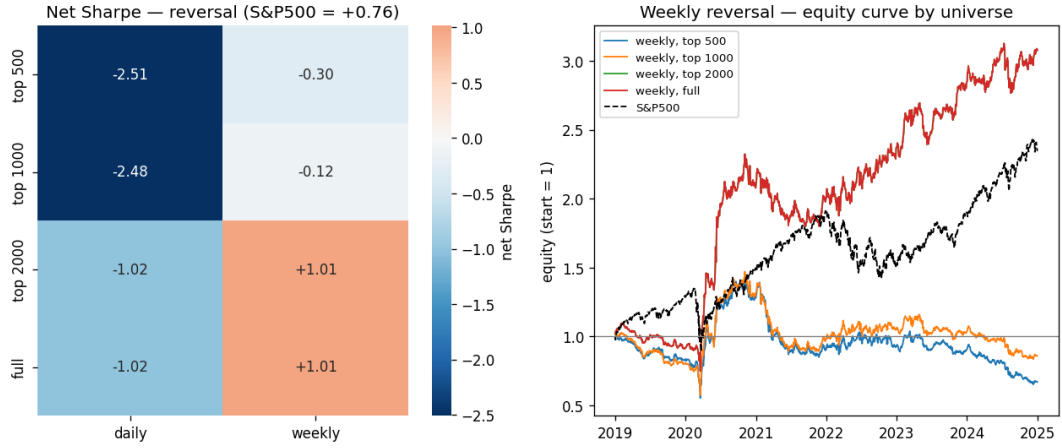


Figure 1: Gross (lighter) and net (darker) Sharpe ratio of weekly-rebalanced reversal across universe definitions. The apparent edge vanishes on a liquid tradable universe.

5 Monthly Cross-Sectional Machine Learning

The daily study established that technical signals alone are insufficient: the dominant gross signal is a micro-cap artifact, and the residual signals are too weak to clear costs on a liquid universe. We therefore pivot to a monthly frequency. This cuts turnover by a factor of ≈ 20 , opens the predictor space to Compustat fundamentals, and makes the prediction target a full calendar month’s return, over which fundamental information is more likely to be priced in.

5.1 Models

Three model classes are trained on the 16-feature vector (5 technical + 11 fundamental) to predict the next-month cross-sectional return rank:

Lasso LassoCV with 5-fold cross-validation (sample capped at 200 k rows for computational tractability); provides a sparse, interpretable linear baseline.

XGBoost 600 trees, maximum depth 5, learning rate 0.05, row and column subsampling 0.8; early stopping against the validation set.

MLP (deep learning) Two hidden layers (64-32) neurons, ReLU, dropout 0.1), trained with Adam for up to 60 epochs with early stopping; inputs standardised per the training set.

An equal-weight (EW) blend of rank-normalised features, oriented by training-set correlations with the target, serves as the baseline. All models are trained on data through 2010-12-31, with 2011–2014 used for hyperparameter selection.

5.2 Full-Universe Results

Table 2 reports out-of-sample performance on the full CRSP monthly universe (L/S decile, 10 bps round-trip, 1% per year borrow). All models extract cross-sectional predictability (IC from 0.079 to 0.093), well above the daily IC of 0.021. However, none beats the S&P 500 Sharpe of +0.77: the full universe includes micro-cap names with extreme monthly return volatility, inflating strategy volatility to ≈ 23 –25% versus 15% for the index. MLP (+0.38) and XGBoost (+0.36) outperform Lasso (+0.20) and EW (+0.17), showing that non-linearity contributes at the monthly horizon, in contrast to the daily case.

Table 2: Monthly L/S decile (test 2015-01 – 2024-11, 10 bps, 1% yr⁻¹ borrow, full CRSP universe). S&P 500 is buy-and-hold.

Model	IC	Sharpe	Ann. Ret.	Ann. Vol	Max DD
Equal-weight	0.079	+0.17	+3.8%	22.8%	−68.5%
Lasso	0.085	+0.20	+5.2%	25.3%	−71.5%
XGBoost	0.094	+0.41	+8.9%	24.9%	−65.8%
MLP	0.089	+0.38	+9.0%	23.9%	−66.8%
S&P 500	—	+0.77	+11.9%	15.4%	−24.8%

5.3 Universe Sweep: Small-Cap Concentration of the Edge

Sorting firms by lagged log-TTM-revenue (a Compustat size proxy that avoids the price/market-cap data restriction) and restricting to the smallest-revenue names, the predictive edge concentrates in small caps. Figure 2 displays XGBoost equity curves for the full, top-500 (largest), and bottom-500 (smallest) revenue universes. Key results:

- **Top-500 (large caps):** IC collapses to ≈ 0.02 ; net Sharpe is negative. Factor anomalies are well-arbitraged in large, heavily-covered firms.
- **Full universe:** IC = 0.093; net Sharpe +0.36.
- **Bottom-500 (small caps):** IC rises to 0.135; net Sharpe +1.12, annualised return +41%. XGBoost beats the S&P 500 (though at high volatility, see Section 8).

This result is consistent with the finding in Gu et al. (2020) and the broader factor literature that anomalies are more pronounced where institutional coverage is lower and pricing efficiency is reduced. The mechanism here is distinct from the daily reversal artifact: positions are held for one full calendar month, which rules out any bid–ask-bounce explanation (Nagel, 2012).

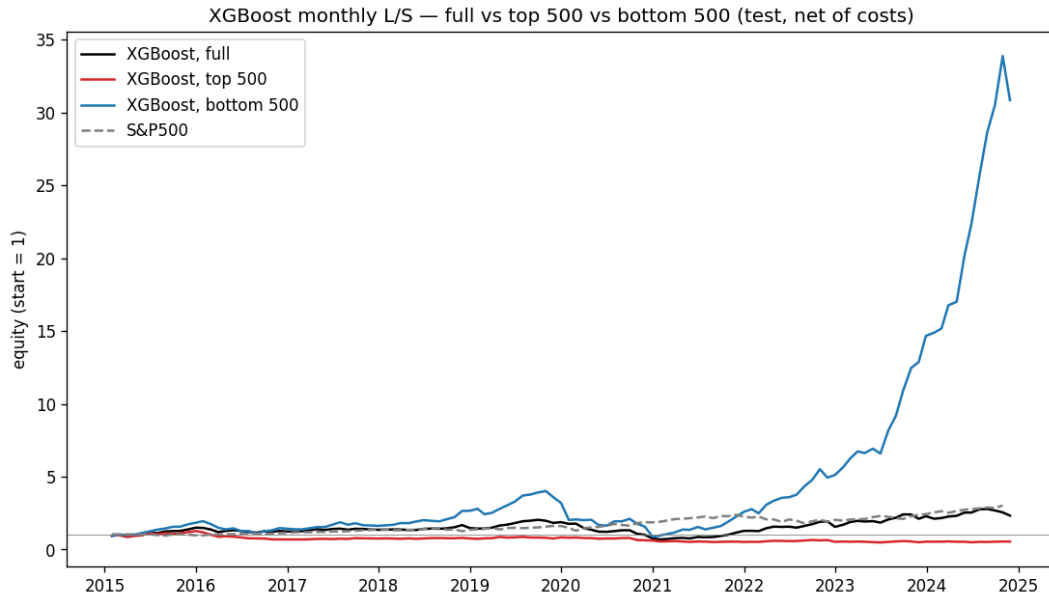


Figure 2: XGBoost monthly L/S equity curves (test 2015–2024, 10 bps) for three universe definitions. The bottom-500 (smallest-revenue) portfolio delivers a net Sharpe of +1.12 vs. S&P +0.77.

5.4 Cost Robustness

Small-cap trading costs exceed the large-cap institutional baseline of 10 bps. Hasbrouck (2009) and Novy-Marx and Velikov (2016) place effective costs for small-cap factor strategies at 50–100 bps. Figure 3 sweeps the round-trip trading cost from 0 to 200 bps for XGBoost on three universes:

- **Bottom-500:** beats S&P 500 (+0.77) up to ≈ 57 bps; at 50 bps the net Sharpe is +0.83, annualised return +30%.
- **Bottom-1 000:** breakeven at ≈ 26 bps.
- **Full universe:** never beats the S&P 500 at any cost level.

The 57-bps breakeven for the bottom-500 falls at the lower end of institutional small-cap cost estimates (50-100bps), suggesting the edge survives only for the cheapest-to-trade end of this range. At retail-level costs (> 100 bps), the advantage disappears entirely. The fact that the bottom-1000 breaks even at lower trading costs hints at the fact that the edge is larger in the smallest names.

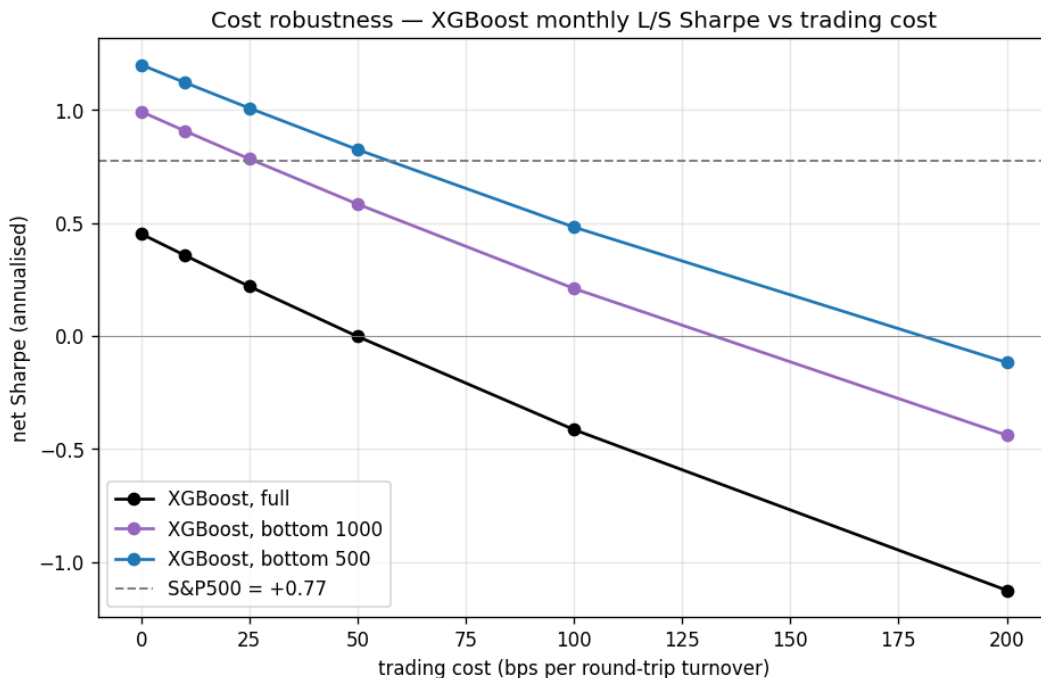


Figure 3: XGBoost monthly L/S net Sharpe as a function of round-trip trading cost. The grey dashed line is the S&P 500 benchmark (+0.77). Bottom-500 breaks even at ≈ 57 bps.

6 Regime Dependence

Table 3 compares the S&P 500 and the long-only low-volatility portfolio across the two sub-periods 2000–2018 and 2019–2024, using daily data.

Over 2000–2018, the S&P 500 achieved a Sharpe of only +0.19 (annualised return +3.7%, volatility 19.0%, maximum drawdown -56.8%) owing to the dot-com crash and the Global Financial Crisis. The long-only low-volatility portfolio delivered +0.76 (annualised return +10.0%), roughly four times the index Sharpe. This is consistent with Baker et al. (2011) and Frazzini and Pedersen (2014). In periods of large drawdowns, low-beta stocks preserve capital and reduce peak-to-trough losses.

Table 3: Regime comparison across two sub-periods on the market-cap top-1 000 tradable universe, net of 10 bps costs (low-vol rebalanced monthly, reversal weekly; S&P 500 buy-and-hold).

Strategy	2000–2018		2019–2024	
	Sharpe	Ann. Ret.	Sharpe	Ann. Ret.
S&P 500	+0.19	+3.7%	+0.76	+15.3%
Low-vol long-only	+0.76	+10.0%	+0.53	+9.0%
Reversal L/S (liq.)	−0.57	−10.6%	−0.12	−2.5%

Over 2019–2024, the relationship reverses. The S&P 500 attained a Sharpe of +0.76 (annualised return +15.3%) as mega-cap technology names drove most of the index’s returns. A low-volatility strategy that underweights these high-beta names systematically underperforms. Reversal L/S returns are negative in both periods on the tradable universe (−0.57 and −0.12), consistent with the liquidity-provision interpretation across regimes.

The monthly test window (2015–2024) overlaps heavily with the mega-cap bull market, a single regime in which the cap-weighted benchmark was unusually hard to beat. Whether the small-cap edge of Section 5.3 would survive or strengthen in a less concentrated regime is an open question that this sample cannot answer.

7 Machine-Learning Meta-Model Comparison

Table 4 summarises out-of-sample IC and net Sharpe for each model class at both frequencies. At the **daily** horizon, all four models, including the LSTM recurrent network, achieve an IC of approximately 0.021 on the tradable top-1 000 universe. Net Sharpes are uniformly negative, ranging from −0.07 (LSTM) to −0.57 (equal-weight blend), so none beats the benchmark, and the near-identical IC indicates that the binding constraint is signal quality, not model sophistication: once the micro-cap bias is removed, no architecture can profitably exploit the residual technical signal.

Table 4: Model comparison across frequencies. Daily: top-1 000 universe, prediction horizon = 1 day, rebalance = weekly, 10 bps, test 2019–2024. Monthly: full CRSP universe, 10 bps, test 2015–2024.

Frequency	Model	IC	Sharpe (net)
Daily	Equal-weight	0.022	−0.57
Daily	XGBoost	0.021	−0.35
Daily	MLP	0.022	−0.27
Daily	LSTM	0.022	−0.07
Monthly	Equal-weight	0.079	+0.17
Monthly	Lasso	0.085	+0.20
Monthly	XGBoost	0.093	+0.36
Monthly	MLP	0.089	+0.38

At the **monthly** horizon the IC spread is narrow (0.079–0.093), but the Sharpe gap is not. MLP and XGBoost each exceed Lasso by +0.16–+0.18 in net Sharpe and EW by +0.19–+0.21. This non-linear advantage is consistent with Gu et al. (2020), who find that neural networks and tree ensembles capture non-linear interactions between firm characteristics that linear models miss. MLP slightly outperforms XGBoost on the full universe (+0.38 vs. +0.36) but *collapses* on the small-cap subsets (bottom-500 Sharpe +0.11, below even Lasso’s +0.71 and far below XGBoost’s +1.12). The MLP’s IC on the restricted sample falls to 0.06 versus 0.14 for XGBoost,

indicating overfitting once the training set shrinks to $\approx 125\text{k}$ rows. Gradient-boosted trees are the more robust choice for the concentrated small-cap strategy.

8 Conclusion

This study documents a clear contrast between daily and monthly cross-sectional return prediction. At the daily frequency, technical signals appear profitable in gross terms, but this profitability is a micro-cap bid–ask-bounce artifact: once a tradable universe is defined, no daily strategy clears realistic transaction costs, and all model classes are equally uninformative. At the monthly frequency, merging CRSP returns with Compustat fundamentals yields cross-sectional predictability. On the full universe, MLP and XGBoost achieve net Sharpe ratios of $+0.38$ and $+0.36$, respectively, below the S&P 500 but well above zero. In the smallest-revenue firm subset, XGBoost delivers a net Sharpe of $+1.12$ that survives trading costs up to ≈ 57 bps, potentially attractive to institutional small-cap investors.

Three points generalise beyond this dataset. The daily horizon’s apparent predictability turned out to be micro-cap contamination. Moving to monthly positions removed the confound and let slower-moving fundamentals into the predictor set, so frequency and liquidity cannot be separated when judging a signal. Model choice mattered only once the problem was rich enough to reward it. Non-linear models earned meaningful Sharpe gains monthly but nothing daily. The benchmark mattered as much as the strategy. The S&P 500 Sharpe quadrupled from 2000–2018 to 2019–2024 on mega-cap concentration, so any diversified systematic strategy faced an unusually hard benchmark in exactly our test window.

Limitations. The bottom-500 small-cap strategy has high annualised volatility ($\approx 37\%$) and extreme drawdowns during COVID-19 ($\approx -78\%$), making it unsuitable without a volatility-targeting overlay (Moreira and Muir, 2017). The ten-year test window covers a single dominant regime. The CCM CUSIP-matching step may exclude a small fraction of firms. The 57-bps breakeven assumes a price-taking investor; at scale, market impact would raise effective costs.

Extensions. Natural next steps stay within the provided data: adding the JKP factor characteristics and the textual signals from the 10-K MD&A sections and earnings-call transcripts (via document embeddings) to the monthly predictor set; applying a long-only volatility-targeting overlay to tame the small-cap drawdown; and a classification reformulation (predicting return-decile bins). The monthly pipeline infrastructure built here is reusable for these.

Code and data availability. All source code, the data pipeline, and this report are in the project repository:

<https://github.com/leo-wun/ML-For-Finance-Project-LeoWunderli-329314-DarioFrey-344524>.

To allow the pipeline to be reproduced without re-downloading from WRDS/CRSP, a single archive of the raw input datasets (Daily/Monthly CRSP, Compustat, Chen–Zimmermann) is available at https://drive.google.com/drive/folders/1h-pifzGcgFoBAj_kZ8oTngdmffxvE6rh?usp=sharing; the repository README gives one-command setup and run instructions.

References

Malcolm Baker, Brendan Bradley, and Jeffrey Wurgler. Benchmarks as limits to arbitrage: Understanding the low-volatility anomaly. *Financial Analysts Journal*, 67(1):40–54, 2011.

Andrew Y. Chen and Tom Zimmermann. Open source cross-sectional asset pricing. *Critical Finance Review*, 11(2):207–264, 2022.

- Eugene F. Fama and Kenneth R. French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015.
- Andrea Frazzini and Lasse Heje Pedersen. Betting against beta. *Journal of Financial Economics*, 111(1):1–25, 2014.
- Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.
- Joel Hasbrouck. Trading costs and returns for U.S. equities: Estimating effective costs from daily data. *The Journal of Finance*, 64(3):1445–1477, 2009.
- Narasimhan Jegadeesh. Evidence of predictable behavior of security returns. *The Journal of Finance*, 45(3):881–898, 1990.
- Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993.
- Bruce N. Lehmann. Fads, martingales, and market efficiency. *The Quarterly Journal of Economics*, 105(1):1–28, 1990.
- Alan Moreira and Tyler Muir. Volatility-managed portfolios. *The Journal of Finance*, 72(4):1611–1644, 2017.
- Stefan Nagel. Evaporating liquidity. *The Review of Financial Studies*, 25(7):2005–2039, 2012.
- Robert Novy-Marx and Mihail Velikov. A taxonomy of anomalies and their trading costs. *The Review of Financial Studies*, 29(1):104–147, 2016.
- Richard G. Sloan. Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3):289–315, 1996.